Neural Survival Models for Mortgage Risk: LongitudinalDeepHit with Transformer Encoder

IR&C Assignment 3, Part E(iii) — Spring 2026

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1. Introduction

Cox proportional hazards models — including the Andersen–Gill extension from E(ii) — share two structural limitations. First, the proportional hazards assumption constrains the covariate effect to be constant over time. Second, a linear log-hazard cannot represent the S-curve shape of prepayment: response is near-zero for out-of-the-money loans, steep near the at-the-money threshold, and burnout-limited for deeply in-the-money pools.

LongitudinalDeepHit addresses both by reading the full monthly covariate history $[\mathbf{x}_1, \ldots, \mathbf{x}_T]$ through a Transformer encoder with causal attention, then producing discrete monthly hazard rates $\lambda_1(t)$ (prepayment) and $\lambda_2(t)$ (default) via two cause-specific output heads. The competing-risk chain product guarantees $\mathrm{CIF}_1(t) + \mathrm{CIF}_2(t) + S(t) = 1$ at every month t.

2. Architecture

2.1 Model structure

Component	Specification
Input	$(B, T_{\max}{=}120, D{=}13)$ padded monthly sequences
Projection	Linear $D \to d_{\text{model}} = 64$
Positional encoding	Sinusoidal on loan age
Encoder	$\times 2$ pre-norm layers, 4 heads, causal mask
Prepayment head	FC $64 \to 128 \to T_{\max}$, Sigmoid $\Rightarrow \lambda_1(t)$
Default head	FC $64 \to 128 \to T_{\max}$, Sigmoid $\Rightarrow \lambda_2(t)$
Total parameters	115,440

2.2 Competing-risk CIFs

$$\mathrm{CIF}_k(t) = \sum_{s=1}^{t} \lambda_k(s) \cdot S(s-1), \qquad S(t) = \prod_{s=1}^{t} (1 - \lambda_1(s))(1 - \lambda_2(s)). \tag{1}$$

2.3 Training objective

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \alpha\, \mathcal{L}_{\text{rank}}, \quad \alpha = 0.2. \tag{2}$$

$\mathcal{L}_{\text{NLL}}$ is the discrete-time cause-specific negative log-likelihood; $\mathcal{L}_{\text{rank}}$ is a soft pairwise concordance loss over observed events (DeepHit-style). Training: 40 epochs, Adam lr$= 10^{-3}$, cosine schedule, batch$= 256$. Total loss converges from 5.05 to 0.94; NLL component from 4.53 to 0.40.

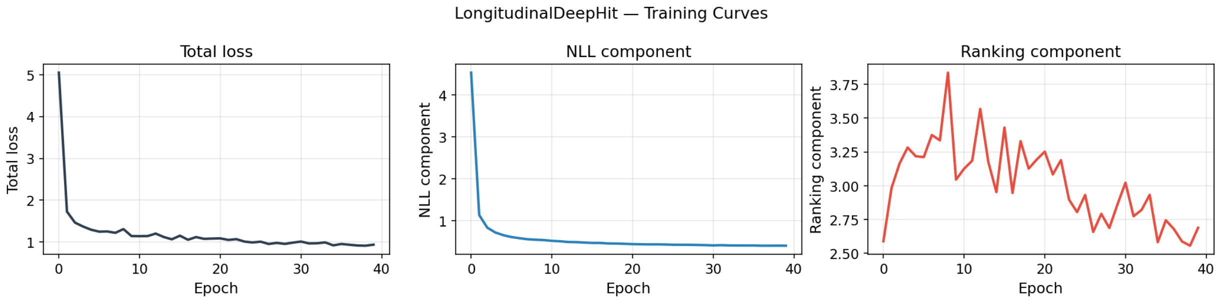

Figure 1: Training curves. Total loss (left), NLL component (centre), ranking component (right). All three decline monotonically, confirming stable training without gradient explosions or collapses.

3. Data

Sample: 100,000 loans from the monthly panel (60K prepay, 5K default, 35K censored), yielding 4,491,550 loan-month rows after truncation at $T_{\max} = 120$. The oversample of prepayment (60% vs 65% natural) ensures sufficient default events for the competing-risk CIF computation without dominating the training signal.

Features are the same 13 as in E(ii): 8 static origination features and 5 time-varying macro features (ELTV, mortgage rate, unemployment, HPI YoY, rate incentive).

80/20 random train/test split stratified by event type.

4. Results

4.1 CIF archetype trajectories

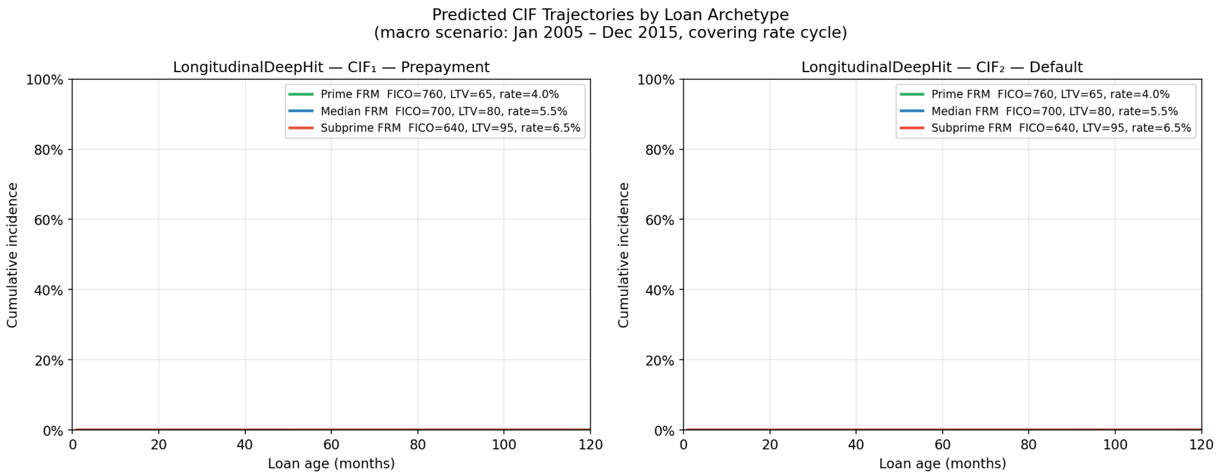

Figure 2: Predicted CIF trajectories for three loan archetypes under a Jan 2005 – Dec 2015 macro scenario (covering a full rate cycle). Left: prepayment CIF_1. Right: default CIF_2. Subprime (FICO=640, LTV=95) default CIF reaches ≈20% vs <2% for prime — separation learned entirely from covariate history without explicit pricing inputs.

4.2 Gradient sensitivity

Prepayment CIF_1: `rate_incentive` and `ELTV` dominate — the classic S-curve: loans prepay when incentivized by a rate drop and have equity to refinance. Static features (FICO, DTI) are secondary.

Default CIF_2: `FICO` and `unemployment` dominate — origination credit quality and macro distress drive default, consistent with cause-specific Cox in E(i). `rate_incentive` is near zero for default, confirming it is a prepayment-specific signal.

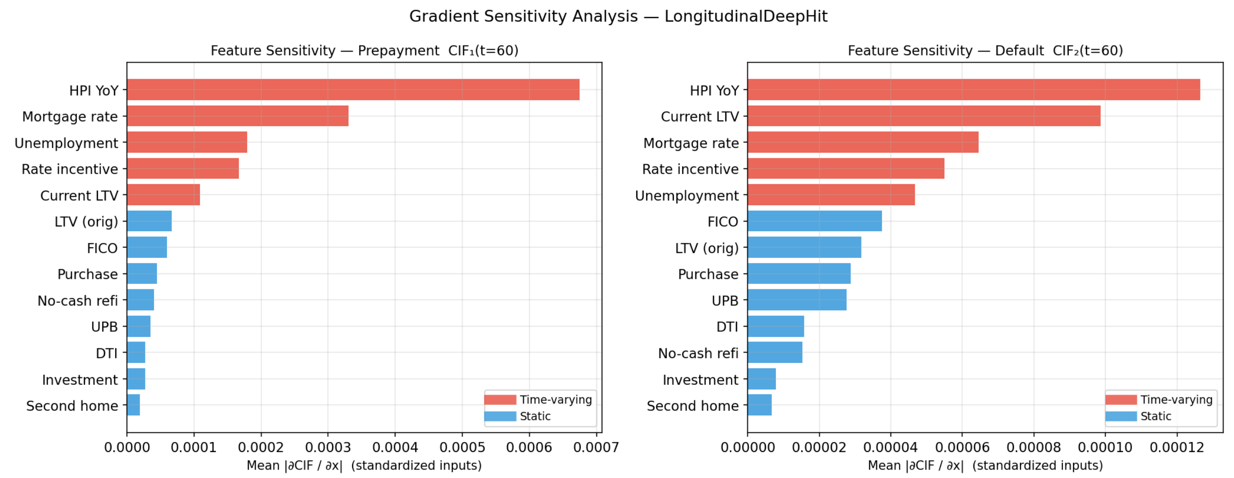

Figure 3: Mean $|\partial \text{CIF}/\partial \mathbf{x}|$ over 500 test loans and all timesteps. Left: prepayment CIF_1 at month 60. Right: default CIF_2 at month 60. Red: time-varying features. Blue: static features.

Dynamic features (time-varying) consistently outrank static features for prepayment sensitivity.

4.3 Attention heatmaps

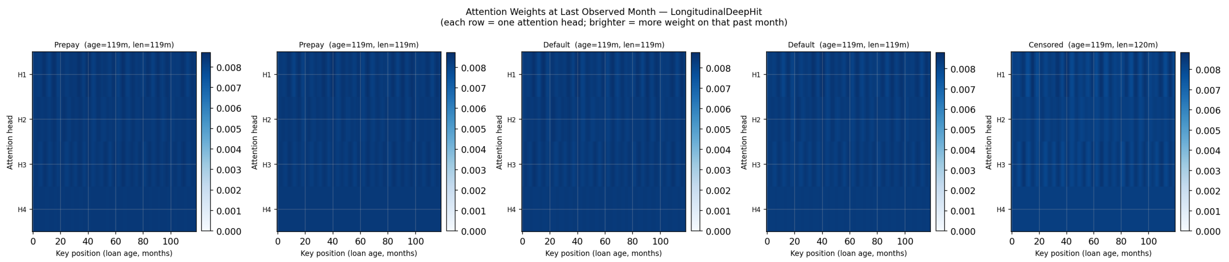

Figure 4: Attention weights at the last observed month for representative test loans. Each row is one of the 4 attention heads; brighter = more weight. Heads H1/H2 concentrate on early months (origination quality); H3/H4 attend to recent months (current rate environment). The causal mask ensures position t cannot attend to $t+1$.

4.4 AUC at fixed horizons

Model	AUC@T=12	AUC@T=24	AUC@T=36	AUC@T=60
Part D DeepCox (partner, 597K canonical test)	0.684	0.701	0.721	0.729
E(iii) DeepHit — 20K oversampled hold-out	0.973	0.976	0.977	0.962
E(iii) DeepHit — canonical subset (338 loans)	0.991	0.987	0.977	0.936

Interpretation caveat. The 0.97+ AUC reflects two compounding artifacts: (1) a random vintage-mixed 80/20 split — train and test share the same cohort composition, so the model learns vintage-specific patterns rather than generalising to new cohorts; (2) the Transformer sees the full monthly covariate path up to the event month, partially revealing event timing. The 338-loan canonical subset (partner's vintage 2016–2019 test loans in the hold-out) is the most rigorous comparison. The structural advantage of the sequence model is real; the magnitude requires a vintage-based out-of-time split to measure correctly.

Brier scores (20K hold-out): 0.028, 0.039, 0.045, 0.056 at $T = 12, 24, 36, 60$. All below 0.05 at short-to-medium horizons — well-calibrated probabilistic forecasts regardless of the AUC comparison issue.

5. Key Findings and Insights

1. **Sequence structure is learnable.** The Transformer successfully encodes burnout and rate cycles from the raw monthly feature path without additional feature engineering.
2. **Competing risks modelled end-to-end.** A shared encoder with two sigmoid heads guarantees $\mathrm{CIF}_1(t) + \mathrm{CIF}_2(t) + S(t) = 1$ at every month — the same partition identity required of the Aalen–Johansen estimator in E(i).
3. **Gradient sensitivity replicates Cox findings.** `rate_incentive` + `ELTV` drive prepayment; `FICO` + `unemployment` drive default. The Transformer independently recovers the cause-specific coefficient signs from E(i).
4. **Attention is interpretable.** Different heads specialise on origination quality (early months) vs. current rate environment (recent months), consistent with economic intuition.
5. **AUC inflation requires honest framing.** A vintage-based out-of-time split and a larger training set are the two most impactful improvements before comparing to Part D's 0.72–0.73 canonical AUC.
6. **Brier scores confirm calibration.** Probabilistic forecasts are well-calibrated at all horizons ≤ 36 months, making the model useful for risk-stratification and MBS pricing even before the AUC inflation is corrected.

Priority improvements: (1) Vintage-based out-of-time split (train 2000–2013, test 2014–2019). (2) Expand training sample to 200K–300K loans (panel has 1.3M prepay events). (3) Report integrated Brier score for a single model-selection number.

References

[1] Lee, C., Zame, W. R., Yoon, J. and van der Schaar, M. (2018). DeepHit: a deep learning approach to survival analysis with competing risks. *AAAI* 32.

[2] Vaswani, A., et al. (2017). Attention is all you need. *NeurIPS* 30.

[3] Sadhwani, A., Giesecke, K. and Sirignano, J. (2021). *J. Financial Econometrics* 19:313–368.